

Weekly Progress Report: Prototypical Networks for Few-Shot Learning

Sayed Mortuza (2311448042) | MD. Farhan (2312703042) | Zawed Bin Tariq (2232811642)
CSE499 Senior Design, North South University | Weekly Update | 29 July 2026

Abstract-This report summarizes the team's weekly study of Prototypical Networks for Few-shot Learning. The work covered episodic training, support and query sets, prototype construction, distance-based prediction, the Conv4 encoder, and the Omniglot and miniImageNet experiments. The main finding is that the model learns a transferable embedding rather than a fixed classifier: prototypes are recomputed from a few labeled examples of unseen classes. This review establishes the theoretical baseline for the next implementation phase.

Index Terms-few-shot learning, meta-learning, metric learning, episodic training, prototypical networks.

I. INTRODUCTION AND OBJECTIVE

The objective for this week was to identify and understand a reliable baseline for the few-shot component of our senior design work. A conventional classifier normally learns a fixed set of output classes and requires many labeled samples. Few-shot learning instead asks a model to recognize novel classes from only one or five labeled examples. We selected the Prototypical Network (ProtoNet) because it provides a simple, reproducible metric-based meta-learning framework [1].

II. WORK COMPLETED

A. Paper and Algorithm Analysis

We completed a structured reading of the paper and mapped its terminology to the training pipeline. In an N-way K-shot episode, N classes are sampled; K labeled examples per class form the support set, while separate examples form the query set. A shared Conv4 encoder maps both sets into one embedding space. The class prototype is the mean of its support embeddings, and a query is classified by a softmax over negative squared Euclidean distances. Query loss is backpropagated through both query and support paths to update the shared encoder. Prototypes are therefore episode-specific calculations, not permanent trainable parameters.

$$c_k = (1 / |S_k|) \sum_{i: y_i = k} f_\varphi(x_i)$$
$$p(y = k | x) = \text{softmax}_k(-||f_\varphi(x) - c_k||^2)$$

B. Dataset and Evaluation Study

We reviewed the two few-shot benchmarks used in the paper. Omniglot contains 1,623 handwritten character classes from 50 alphabets, with 20 samples per class [2]. miniImageNet contains 100 real-world image classes with 600 images per class and uses a 64/16/20 class split for training, validation, and testing. The most important evaluation rule is that test classes are disjoint from training classes. We also clarified that 1-shot means one labeled support image per class in an episode, not one image for the entire training process.

III. WEEKLY STATUS

The literature-review milestone is complete. Deliverables produced this week include a structured paper summary, definitions of the core terms, a benchmark comparison, the mathematical pipeline, and an equation-to-code map for PyTorch. Baseline implementation and empirical reproduction have not yet been completed; therefore, the accuracies below are the verified results reported by the original paper, not results from our own experiments.

IV. KEY FINDINGS

ProtoNet uses four identical convolutional blocks, each containing a 3 x 3 convolution with 64 filters, batch normalization, ReLU, and 2 x 2 max pooling. Its contribution is not a complex backbone; it is the combination of episodic learning and a dynamic nearest-prototype classifier. The paper

reported 98.8% and 99.7% accuracy for 5-way 1-shot and 5-shot Omniglot. On the harder miniImageNet benchmark, the corresponding results were 49.42 +/- 0.78% and 68.20 +/- 0.66% [1]. Five-shot learning performs better because averaging multiple support embeddings reduces sample noise. The experiments also showed that Euclidean distance was better suited to mean prototypes than cosine distance, and that harder higher-way training can produce a more discriminative embedding.

V. LIMITATIONS AND DESIGN RELEVANCE

A basic ProtoNet represents every class with one mean vector and therefore assumes a compact, approximately single-cluster class distribution. Multimodal classes, mislabeled or unusual support samples, and domain shift can move the prototype away from representative examples. Performance is also sensitive to episode configuration, distance choice, and encoder quality. These limitations make ProtoNet useful both as a baseline and as a starting point for possible extensions such as multiple prototypes, robust prototype estimation, or feature alignment.

VI. RISKS AND VALIDATION CONTROLS

The main implementation risk is making an unfair comparison because preprocessing, class splits, episode sampling, query count, and random seeds can change accuracy. To control this, we will freeze the data split, image transformations, seeds, way, shot, and query settings, optimizer, and evaluation episode count before testing. Every deviation from [1] will be logged, and results will be reported as mean accuracy with a 95% confidence interval.

VII. NEXT-WEEK PLAN

The next milestone is a clean PyTorch reproduction. We will: (1) implement the four-block encoder and episodic sampler; (2) run 5-way 1-shot and 5-shot Omniglot experiments; (3) verify support, query, prototype, distance, and label tensor shapes; (4) record loss and episode accuracy; and (5) compare our configuration and results with the paper. Code review and experiment logs will be shared so that all team members understand the complete pipeline.

VIII. CONCLUSION

This week produced a complete conceptual understanding of ProtoNet, its datasets, training logic, reported results, and limitations. The team is now prepared to move from literature review to an independently verified baseline implementation.

REFERENCES

- [1] J. Snell, K. Swersky, and R. Zemel, "Prototypical Networks for Few-shot Learning," in Proc. 31st Conf. Neural Information Processing Systems, 2017.
- [2] B. M. Lake, R. Salakhutdinov, and J. B. Tenenbaum, "Human-level concept learning through probabilistic program induction," Science, vol. 350, no. 6266, pp. 1332-1338, 2015.
- [3] O. Vinyals et al., "Matching Networks for One Shot Learning," in Proc. Advances in Neural Information Processing Systems, 2016.